

Artificial Intelligence Course 1

WELCOME to A.I.



Games Change Everything Module 4

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Fundamentals of Best Practices in A.I.



Games Change Everything Module 04

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A close up of the latest BabyX's eyes

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Module 4 demonstrates games and A.I.
are accomplishing interesting things

◀ ● ▶
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Baby X by Mark Sagar University of
Auckland

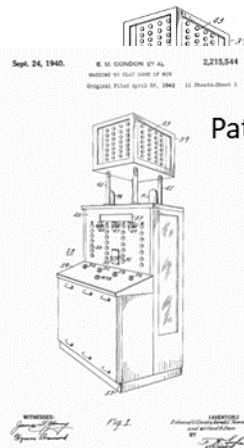
Jeremy Norman's
HistoryofInformation.com

Outline Expanded Map

A: Pittsburgh, Pennsylvania, United States, B: Queens, New York, United States

NIMATRON: An Early Electromechanical Machine to Play Chess

1940 CE



Patent

For the Westinghouse Pavilion at the 1939 World's Fair, physicist Edward Condon and the Westinghouse Electric Company patented an electromechanical machine that could play chess. The machine was designed as a promotional strategy for the company's new line of vacuum tube electronic calculators. It was first demonstrated in April 1940, after which it was shown at the World's Fair. They received U.S. Patent 2,215,544 on September 24, 1940. The machine was built and reproduced with the patent. Other drawings in the patent show the machine playing 100,000 games at the fair. It was a success, demonstrating that possibility. When it was first shown, it was sent its opponent with a token of appreciation.

In 1939 Westinghouse produced a 55 minute promotional film: *The Middleton Family at the New York World's Fair*, it may be viewed

The Middleton Family at the New York World's Fair

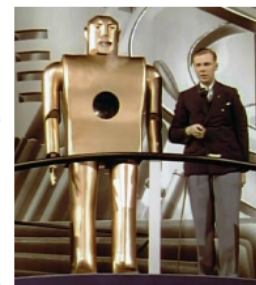
By Andrew F. Wood

"The Middleton Family at the New York World's Fair" culminated a broader campaign of mass-market magazine ads and radio spots designed to showcase the Westinghouse Electric and Manufacturing Company's pavilion at the 1939-40 international exhibition. Though more a time capsule of corporate propaganda than cinematic excellence, the film still offers a fascinating glimpse into commercial strategies employed to confront (and, to a certain extent, exploit) America's late Depression-era anxieties about technology, automation, and the Red Menace.

The Middletons represent Westinghouse's vision of a typical American family. Father has worked hard for his little piece of middle class comfort, and he sees no reason that the future ought not hold even better prospects for his kids. Mother is a bit more high-strung but no less assured in her family and their prospects. Their teenage son Bud, though, is less confident. Ostensibly irreverent, Bud's wisecracks disguise a fear that the world of tomorrow offers his generation little hope, only declining employment and shrinking opportunity.

The threesome has journeyed from Indiana to New York to attend the Fair -- and since they are staying with Grandma, they anticipate a reunion with Bud's older sister Babes. Unfortunately they soon discover that the young woman has begun to affect an overly worldly attitude, one coinciding with her participation in a love triangle pitting two markedly different suitors: Jim Treadway, the hometown hero who works for Westinghouse and dispenses information to Fair visitors, and Nicholas Mallard, a sleazy looking fellow who teaches art theory at Babes' college.

Amid this romantic tension, the Middletons tour the Westinghouse Building, stopping first at the Time Capsule scheduled to be unearthed 5,000 years hence. Leading the tour, Jim describes how committees and authorities, the "brains of the world," have worked carefully and well to ensure that the Westinghouse message to the future will endure. Later he shows off more exhibits, including a television display, the Playground of Science, the Junior Science Hall, and other highlights.



Electro, the Westinghouse "Moto-Man," was a popular attraction at the fair. Courtesy Library of Congress Collection.

At the Playground of Science (where visitors can "play with electricity"), Jim explains his choice to forgo his childhood dream of playing football. It is so much better, he proposes, to work with Westinghouse men dedicated to building the world of tomorrow. Father enthuses: "One can't help marveling at all the new things they're inventing every day." For Jim, this statement offers an opportunity to gird his optimistic rhetoric with an economic appeal: "A company can't stand still under the American system of private enterprise. It has two ways to go: Back or forward."

Mother and Grandma take their own tour, stopping to gaze upon the "Battle of the Centuries." The exhibit features two characters, "Mrs. Drudge" and "Mrs. Modern" who compete in a dishwashing contest. Predictably Mrs. Drudge is splashing through a pile of dirty dishes, cleaning them by hand. Dependent upon her own individualized labor, Mrs. Drudge signifies the past. Standing nearby, serene, Mrs. Modern resides already in the future, a world of gleaming electrical devices epitomized by her Westinghouse dishwasher. This contest demonstrates a similar message to the one learned earlier by Bud: mechanized efficiency triumphs over brute labor.

Bud and Jim catch up with Babes and Nicholas, and



<https://en.wikipedia.org/wiki/Nimatron>

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Presented at the NY World's Fair 1940

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Earliest Computer Game 1940
The Nimatron

Game theory is the mathematical study of interaction among independent, self-interested agents. The audience for game theory has grown dramatically in recent years, and now spans disciplines as diverse as political science, biology, psychology, economics, linguistics, sociology and computer science—among others. What has been missing is a relatively short introduction to the field covering the common basis that anyone with a professional interest in game theory is likely to require. Such a text would minimize notation, ruthlessly focus on essentials, and yet not sacrifice rigor. This Synthesis Lecture aims to fill this gap by providing a concise and accessible introduction to the field. It covers the main classes of games, their representations, and the main concepts used to analyze them. “This introduction is just what a growing multidisciplinary audience needs: it is concise, authoritative, up to date, and clear on the important conceptual issues.” —Robert Stalnaker, MIT, Linguistics and Philosophy

Kevin Leyton-Brown University of British Columbia, Vancouver, BC, Canada

<http://cs.ubc.ca/~kevinlb>

Yoav Shoham Stanford University, Palo Alto, CA, USA

<http://cs.stanford.edu/~shoham>

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A game is a structured form of play, usually undertaken for entertainment or fun, and sometimes used as an educational tool. Games are distinct from work, which is usually carried out for remuneration, and from art, which is more often an expression of aesthetic or ideological elements.

Perhaps we are expanding the environment known as “games” and moving into a more mature form of engagement whereby exploration, imagination and hand’s-on innovation have an appropriate role in work/discovery. Just maybe it is possible to channel the creativity process by exploiting idea Simulation in safe trial and error.

**But now, is all of this changing?
What would you call it?**

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<https://www.youtube.com/watch?v=4wQ4ZmPuy9w>

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Games have been serious entertainment



Games are becoming serious scientific
and business development tools

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CMS.611, Fall 2014

Creating Video Games

Rik Eberhardt Guest Scot Osterweil, MIT Game Lab

**Session 15: Freedoms of Play by Scot Osterweil
(MIT Game Lab);
Project 4 Playtesting; Project 4
Status Report**

MITOPENCOURSEWARE
MASSACHUSETTS INSTITUTE OF TECHNOLOGY



Play is exploration constructing our understanding of the world
Primitive sense of modelling the world – describing the world mathematically

©

<https://www.youtube.com/watch?v=s8At7cnDeIQ>



Guest Scot Osterweil, MIT Game Lab

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What is PLAY?

What is a game? (Bernard Suits)

Freedom to fail, identify, freedom of effort, you can't make people play

- Playing a game is the voluntary attempt to overcome unnecessary obstacles.
- No one is asking you explicitly to move in such potentially cheesy ways. But...
 - It can be challenging
 - It is engaging with flow
 - It is definitely fun if "designed right"
- To me, that's definitely clubbing!

Play is are intrinsic, personal and internal



Oldest Known Game is equal to the oldest known writing 6,000 years ago

©

Learning is happening through play

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Bernard Suits - voluntarily
overcoming unnecessary obstacles

WHAT IS A GAME?*

BERNARD SUITS
University of Waterloo

By means of a critical examination of a number of theses as to the nature of game-playing, the following definition is advanced: To play a game is to engage in activity directed toward bringing about a specific state of affairs, using only means permitted by specific rules, where the means permitted by the rules are more limited in scope than they would be in the absence of the rules, and where the sole reason for accepting such limitation is to make possible such activity.

Prompted by the current interest of social and behavioral scientists in games, and encouraged by the modest belief that it is not demonstrably impossible for philosophers to say something of interest to scientists, I propose to formulate a definition of game-playing.

1. Game-Playing as the Selection of Inefficient Means. Mindful of the ancient canon that the quest for knowledge obliges us to proceed from what is knowable to us to what is knowable in itself, I shall begin with the commonplace that playing games is different from working. Games, therefore, might be expected to be what work, in some salient respect, is not. Let me now baldly characterize work as "technical activity," by which I mean activity in which an agent (as *rational worker*) seeks to employ the most efficient available means for reaching a desired goal. Since games, too, evidently have goals, and since means are evidently employed for their attainment, the possibility suggests itself that games differ from technical activities in that the means employed in games are not the most efficient. Let us say, then, that games are goal-directed activities in which inefficient means are intentionally (or rationally) chosen. For example, in racing games one voluntarily goes all around the track in an effort to arrive at the finish line instead of "sensibly" cutting straight across the infield.

The following considerations, however, seem to cast doubt on this proposal. The goal of a game, we may say, is winning the game. Let us take an example. In poker I am a winner if I have more money when I stop playing than I had when I started. But suppose that one of the other players, in the course of the game, repays me a debt of a hundred dollars, or suppose I hit another player on the head and take all of his money from him. Then, although I have not won a single hand all evening, am I nevertheless a winner? Clearly not, since I didn't increase my money as a consequence of playing poker. In order to be a winner, a sign and product of which is, to be sure, the gaining of money, certain conditions must be met which are not met by the collection of a debt or by felonious assault. These conditions are the rules of poker, which tell us what we can and what we cannot do with the cards and the money. Winning at poker consists in increasing one's money by using only those means permitted by the rules, although mere obedience to the rules does not by

* Received June, 1966.

The essential elements of a game are players, actions, information, strategies, payoffs and equilibria (a state of balance between opposing forces or actions). The sets of players, strategies and payoffs combine to give us the rules of the game states. Fiona Hayes in her paper, "Is Game Theory useful for the analysis and understanding of decision making in economic settings?"

When a few firms dominate a market, when countries have to agree on a trade policy or when a government deregulates a market are just three examples of when a player has to consider other agents' reactions and expectation regarding their own decisions, i.e. strategic interaction.

One of the main contributions from von Neumann's and Morgenstern's book, *The Theory of Games and Economic Behavior* (1944), is their analysis of two-person zero-sum games. In a zero-sum game, the gains of one player are equal to the losses of the other player. The amount of "winnable" goods is fixed i.e. the sum of the gains and the losses equals zero. In non-zero-sum games, the total amount gained is variable. Both players may win (or lose). When they can both win by cooperating in some way, it can be said that their cooperation creates a synergy (*Principia Cybernetica* Web © 1993).

As early as 1928, von Neumann introduced the minimax solution for a two-person zero-sum games. According to the minimax solution, each player tries to maximize his gain in an outcome that is disadvantageous to him (where the worst outcome is determined by his opponent's choice of strategy). With such a strategy, each player can guarantee himself a minimum gain. However, the players' choices of strategy may or may not be consistent with each other's. Even so, von Neumann showed that there is always a minimax solution, i.e. a consistent solution, if "mixed" strategies are introduced. A mixed strategy is a probability distribution of a player's available strategies, whereby a player is assumed to choose a certain "pure" strategy with some probability.

In his doctoral dissertation entitled, *Noncooperative Games*, John Nash formulated a universal solution concept with an arbitrary number of players and arbitrary preferences (not just two-person zero-sum games). This solution concept

later came to be known as the Nash equilibrium. In a Nash equilibrium, all of the players' expectations are fulfilled and their chosen strategies are optimal. Nash proposed two interpretations of the equilibrium concept: one based on rationality and the other on statistical populations that is used in evolutionary games.

For the rational interpretation, the players are perceived as rational (utility maximizers). They have complete information

about the other players' situations.

Reinhard Selten introduced the requirement of subgame perfection, i.e. only credible threats should be taken into account. If a bluff is called and the threat is not carried out, it is not a viable Nash equilibrium. Selten also introduced another refinement usually called the "trembling hand" equilibrium. Each player presupposes a small probability that a mistake will occur: someone's hand will tremble.

John C. Harsanyi provided a way to analyze games when players have no or incomplete information about the other players. This presumption actually best reflects many strategic interactions in the real world.

Harsanyi postulated that every player is one of several "types," and each type corresponds to a set of possible preferences for the player and (subjective) probability distributions over the

other players' types. Harsanyi showed that for every game with incomplete information, there is an equivalent game with complete information. Thus, he transformed games with incomplete information into ones with imperfect information that could be handled with standard methods. (Source: press release for The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel (1994)) This incomplete information can also be greatly enhanced by simulations to look at the uncertainties and possible interactions and resulting outcomes.

In this supplement, the primary study looks at competing in a standard-based industry. These lessons are punctuated with game theory's insight on the players' behavior and the resulting effect on our "business culture" i.e., ethical stances. The game within the game so to speak.—MKC

The essence of game theory

We believe that it is necessary to know as much as possible about the behavior of the individual and about the simplest forms of exchange.—John von Neumann and Oskar Morgenstern

The distinction between cooperative and non-cooperative games is unrelated to the mathematical description by means of pure strategies and pay-off functions of a game. Rather it depends on the possibility or impossibility of coalitions, communication and side-payments.

The main mathematical result is the proof of the existence in any game of at least one equilibrium point....As an illustration, a simple three-man poker model is included. —from the Abstract of John F. Nash, Jr.'s doctoral dissertation (26 May 1950)

Plans based on the assumption that average conditions will occur are wrong on average.

—Sten Savage, senior research engineer and director of industrial affiliates, Stanford University (CA)

about the structure of the game, including all of the players' preferences regarding possible outcomes, where this information is common knowledge. As a result, they can compute each others' strategic alternatives and preferences. They can also compute each others' optimal choice of strategy for each set of expectations. If all the players expect the same Nash equilibrium, then there are no incentives for anyone to change his or her strategy.

The Nash equilibrium has become a standard tool in almost all areas of economic theory. The most obvious is perhaps the study of competition between firms. But it does have problems. If a game has several Nash equilibria, the equilibrium criterion cannot be used immediately to predict the outcome of the game. Also, in terms of player rationality, the equilibrium concept presumes that each player has complete information

APRIL/MAY 2003

<https://www.jstor.org/stable/186102>

Both Articles are great

Online - LIVE

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The PDF article is a simple download

The Video Game Industry

The video game industry is larger (in terms of revenue) than the film and music industries combined. In the last 20 years, the video gaming industry drove forward huge advances in GPUs (graphical processing units), used to do the matrix math needed for rendering graphics. Fortunately, these are exactly the type of computations needed for deep learning. These advances in GPU technology are a key part of why neural networks are proving so much more powerful now than they did a few decades ago. Training a deep learning model without a GPU would be painfully slow in most cases.

With AlphaStar it is becoming clear Games play a pivotal role in providing new platforms to better understand more sophisticated pursuits in A.I., Learning, The Sciences, The Arts & Humanities, Business, Education, etc.

©

Why Would Games Change Everything?

Completely
Harnesses
A Better
Way of Doing

Own the
Horsepower
To Enable
Massive Change

The Future
Tool Sets
Are
Game Based

The Industry
set in place
The Future
Pipelines

Great Enablers

Raytracing A.I.

GPU's & FPGA's

Presence Tech

Distribution

©

What Have Games Done For Us?

Pushed the Envelope on:

- Real-Time Distribution Networking with Security
- Real-time Graphics Processing and GPU's
- Scalability across Devices and metrics
- Interactivity, Augmented Reality, Token Handling
- Game Engines that increase life-like process operation
- Tournament Play and Scoring Technology
- Sound/Audio/Music and Voice Recognition
- Gesture and Haptic Multi-modal Controls
- News/Live Action Play/Record
- Games as Modelling and Learning Environments
- High Definition and Immersive Video Technology VR
- Facial and Image Recognition
- Communication Technology

©

Games have driven high performance
simulation

High Performance GPU's have
provided remarkable horsepower

Evans and Sutherland

The company was founded in 1968 by David C. Evans and Ivan Sutherland, professors in the Computer Science Department at the University of Utah, who were pioneers in computer graphics technology.^[2] Most of the employees were active or former students, and included Jim Clark, who started Silicon Graphics, Ed Catmull, co-founder of Pixar, John Warnock of Adobe, and Scott P. Hunter of Oracle.

In the early 1970s they purchased General Electric's flight simulator division and formed a partnership with Rediffusion Simulation, a UK-based **flight simulator** company, to design and build digital flight simulators. For the next three decades this was E&S's primary market, delivering display systems with enough brightness to light up a simulator cockpit to daytime light levels. These simulators were used for training in in-flight refueling, carrier landing, AWACS, and B52 CAORF (Computer Aided Operations Research Facility), AND CDRS

E&S produced the **LDS-1 (Line Drawing System-1), Picture System 1, 2 and PS300 series. These unique "calligraphic" (analog vector drawing) color displays had depth cueing and could draw large wireframe models and manipulate (rotate, shift, zoom) them in real time.** They were used both in chemistry by pharmaceutical companies to visualize large molecules such as enzymes or polynucleotides, and by aerospace companies, such as Boeing, McDonnell-Douglas and others, to design aircraft. The end of the Picture System line came in the late 80s, when raster devices on workstations could render anti-aliased lines faster. Wikipedia

The MindSet virtual set system was created to address the needs of the broadcast video market.^[6]

E&S in creation of the Project Genesis simulation sequence in *Star Trek II: The Wrath of Khan*

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The famous Teapot

©

The most famous Teapot in Graphics

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One of Earliest Light Pen On screen
drawing "Sketchpad" System

Computer graphics terminals—A backward look

by CARL MACHOVER
Information Displays, Inc.
Mt. Kisco, New York

INTRODUCTION

Five years ago, the price of admission into Interactive Computer Graphics was spending about \$50,000 or more for the Graphics Terminal and associated hardware, plus writing almost all of the applications software, as well as much of the basic software. And, only about a dozen suppliers offered commercial equipment.

Today, the price of admission has dropped dramatically. Graphics Terminals can be purchased for less than \$10,000. Some turnkey applications software packages are available. And the buyer can choose from among more than 35 hardware and system suppliers, offering over 60 different models.

Some aspects of graphics terminal performance have not changed materially over the past five years. For example, maximum screen data content (number of flicker-free points, characters and lines) has remained almost constant for refresh displays. However, there have been significant advances in other areas, such as intelligent (minicomputer-based) terminals, low cost terminals, color displays, specialized hardware function devices, graphic tablets, and the use of storage tubes and digital TV to increase screen content.

SUPPLIERS

In my FJOC Paper about five years ago,¹ I listed sixteen manufacturers of commercially available CRT graphic terminals. A comparison between that list and an updated version compiled from Computer Display Review,² and Modern Data Systems,³ is given in Table I.

The number of suppliers has more than doubled in the past five years. During the past five years, several companies, such as Adage, IBM, and IDI, have offered upgraded versions of earlier systems. Probably the most widely used graphic terminals over the past five years were the IBM 2250 Series units. Originally intro-

duced in Spring 1964 with the IBM 360 Series computer, three additional versions were subsequently offered.

Two companies (Stromberg Carlson and Philco-Ford) have essentially withdrawn from the commercial field. Several companies do not appear in either list, because either they introduced and then withdrew products in the intervening years, or they introduced products and were then merged into another company. For example, Coming Data Systems and Graphics Display Ltd (England) both introduced low cost graphic terminals several years ago and then either formally, or informally, withdrew them from the market a year or so later. Computer Displays, Inc. introduced the first low cost graphic terminal (using a storage tube) about four years ago, and then merged into Adage about a year ago, losing its corporate identity.

I will not be surprised if there are other changes by the time this is published in May 1972 . . . new products and suppliers, mergers, or product withdrawals. For example, plasma panels and liquid crystal panels with associated displays are just now becoming commercially available from Owens-Illinois and Optel, respectively.

I estimate that the companies listed have each spent in the range of \$250,000 to \$1,000,000 to bring these commercial products into the market place. Perhaps, then, some \$50,000,000 has been invested in these terminals, whose current installed value is about equal to that investment. Certainly, graphic terminal business is not a "get-rich-quick" scheme!

TERMINAL CONFIGURATIONS

Intelligent terminal

Five years ago, most terminals consisted of a display generator (with digital logic and some analog function generators) and a refreshed CRT. Only one system used a storage tube (the BBN Teleputer System), and only two systems included their own computers (DEC

430

Departments | Geek Life | History

27 Oct 2017 | 16:00 GMT

Behold, the World's Most Famous Teapot

Used in pioneering computer-graphics research, the Utah teapot has made cameos in Pixar's *Toy Story*, Disney's *Beauty and the Beast*, and "The Simpsons"

By David C. Brock



Photo: Mark Richards/Computer History Museum

Martin Newell was worried about his Ph.D. research as he sat down to tea with his wife one day in 1974. His work, at the University of Utah, was in the relatively new field of computer graphics. Some years earlier, the computer scientists David Evans and Ivan Sutherland had come to Utah, transforming the university into the world's leading center for computer graphics research.

Evans, who had deep family connections to Utah, was a rising star when the University of Utah, in Salt Lake City, had recruited him to start its computer science department in 1965. At the time, he'd been on the faculty of the University of California, Berkeley, and was one of the principal investigators on Project Genie [PDF]. This pathbreaking effort on time-shared computing was supported by the U.S. Defense Department's Advanced Research Projects Administration, or ARPA.

In 1968, Evans lured Ivan Sutherland away from a tenured professorship at Harvard to join him in Utah at the computer science department, as well as at a new company the pair would form, Evans & Sutherland. Sutherland was a pioneer in computer graphics, beginning with his development of the famed Sketchpad software for his 1963 Ph.D. at MIT; his thesis advisor was Claude Shannon. Sketchpad is widely viewed as the first computer-aided design program. In this 1994 talk, Sutherland recounts how he created Sketchpad:

©

Industrial Graphics and Games shared
much the same tools

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Games provided a great deal of
the development for Industrial use

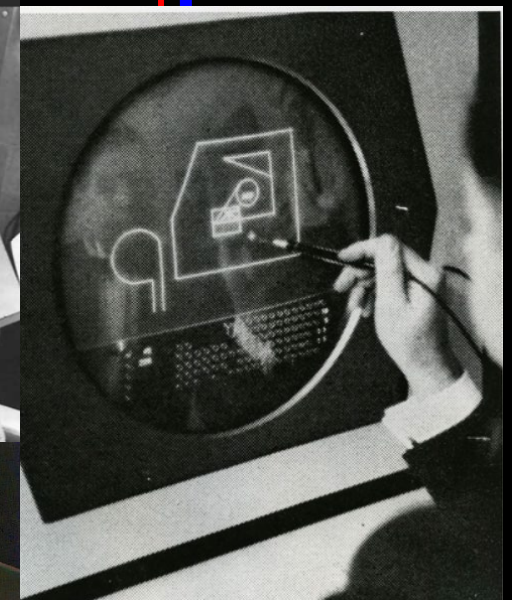
"glass" C-5M instrument panel

A **glass cockpit** is an aircraft cockpit that features electronic (digital) flight instrument displays, typically large LCD screens, rather than the traditional style of analog dials and gauges.^[2] While a traditional cockpit relies on numerous mechanical gauges (nicknamed "steam gauges" within aviation circles) to display information, a glass cockpit uses several multi-function displays driven by flight management systems, that can be adjusted to display flight information as needed. This simplifies aircraft operation and navigation and allows pilots to focus only on the most pertinent information. They are also popular with airline companies as they usually eliminate the need for a flight engineer, saving costs. In recent years the technology has also become widely available in small aircraft. As aircraft displays have modernized, the sensors that feed them have modernized as well. Traditional gyroscopic flight instruments have been replaced by electronic attitude and heading reference systems (AHRS) and air data computers (ADCs), improving reliability and reducing cost and maintenance. GPS receivers are usually integrated into glass cockpits.

Glass cockpit in a Cirrus SR22

©

In 1963, Ivan Sutherland engineered a revolution in computer graphics with his highly-interactive program **Sketchpad**. It enabled users to design and draw in real time directly on the computer display, using a light pen. Among other innovative features, Sketchpad allowed users to manipulate, duplicate, store, and recall drawings for future use. The genesis of a multibillion-dollar industry, Sketchpad pioneered the way for human-computer interaction; it was the forerunner of the graphical user interface (GUI) programs found today in all computers, gaming devices, MP3 players, and smartphones. Sutherland is widely regarded as the "father of computer graphics". Wikipedia



CAD

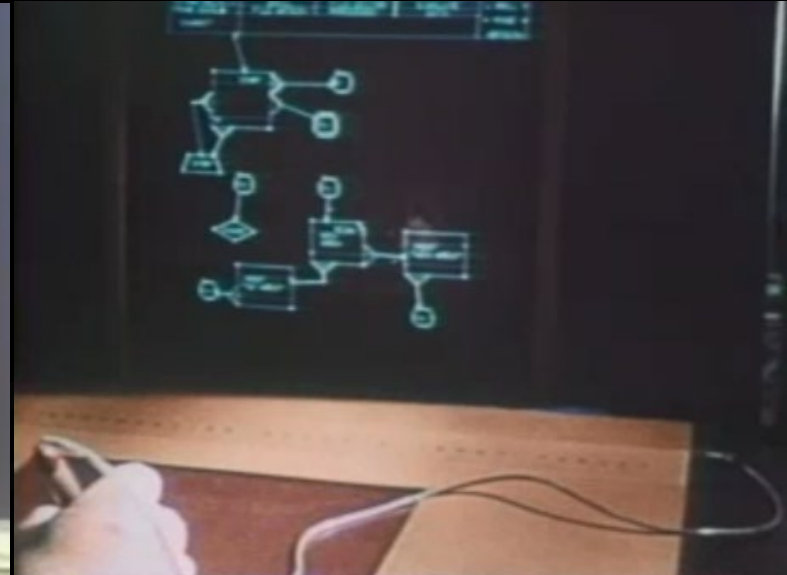


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Sketchpad inspired Douglas Engelbart to design and develop ON-Line System at the Augmentation Research Center (ARC) at the Stanford Research Institute (SRI) during the 1960s.

©

<https://archive.org/details/AlanKeyD1987>

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Alan Kay represents the exciting years at Xerox PARC

Online - LIVE

Presence Technology “Virtual Reality”

Virtual reality (VR) is a simulated experience that can be similar to or completely different from the real world. Applications of virtual reality can include entertainment (i.e. video games) and educational purposes (i.e. medical or military training). Other, distinct types of VR style technology include augmented reality and mixed reality, sometimes referred to as extended reality or XR.^[1]

Currently standard virtual reality systems use either virtual reality headsets or multi-projected environments to generate realistic images, sounds and other sensations that simulate a user's physical presence in a virtual environment. A person using virtual reality equipment is able to look around the artificial world, move around in it, and interact with virtual features or items. The effect is commonly created by VR headsets consisting of a head-mounted display with a small screen in front of the eyes but can also be created through specially designed rooms with multiple large screens. Virtual reality typically incorporates auditory and video feedback but may also allow other types of sensory and force feedback through haptic technology. Wikipedia

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Courtesy Sawyer's Photo Finishing and Gruber-Sawyer



Stereo Viewing

The good Ol' ViewMaster

Courtesy Virtuix VR Omni Treadmill



Complete Treadmill VR Experience

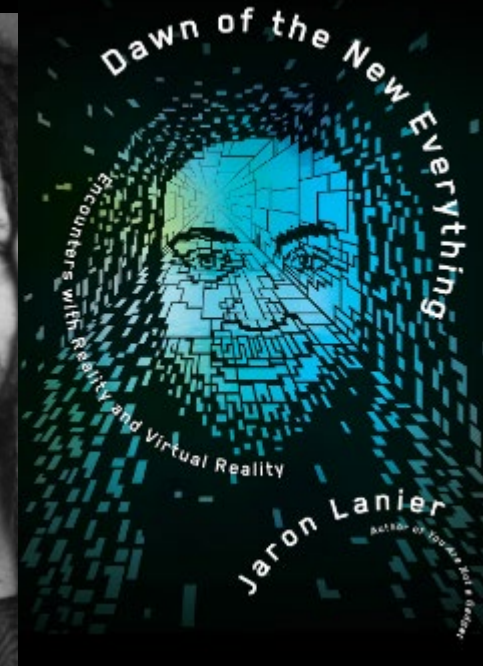
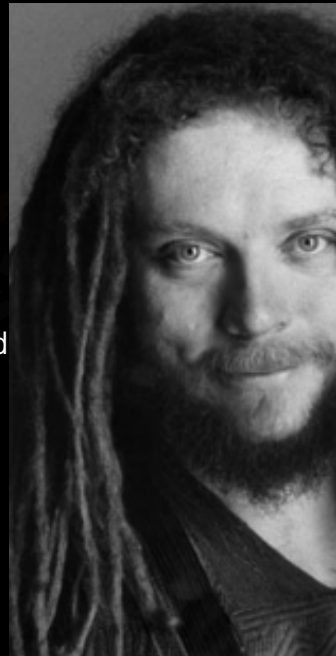
The Virtual Reality Experience has taken a long time to launch but
It may end up being one of the most ubiquitous environments for collaboration.

©



First Head Mounted display

VR technology is based on a row of inventions, such as the stereoscope, which allowed for the first three-dimensional illustration of images in the 19th century. The story of VR in a closer sense begins in 1968 with the "Sword of Damocles", which was developed by Ivan Edward Sutherland and his student Bob Sproul. This was the first head mounted display (HMD) and is therefore a predecessor to the modern VR headsets. The nickname "Sword of Damocles" was given to the device, because it was mounted on the ceiling and looked like it. Sutherland developed the first graphic program capable of 3D drawings on computers. **"Sword of Damocles"** The first VR phase ran between 1982 and 1992. The developments by Jaron Lanier and his VR company laid the foundation for this. Building and programming went hand in hand back then. Because the programs, VR headsets and input devices did not exist. The VR pioneers had to do everything themselves and find out what is possible.



Jaron Lanier (VPL) "Father of Virtual Reality"

©

SCIENTIFIC AMERICAN

OCTOBER 1987
\$2.50

The next revolution in computers, the subject of this issue, will see power increase tenfold in 10 years while networks and advanced interfaces transform computing into a universal intellectual utility.



Wind Glove gives a computer user the feel of the screen; the image of the hand is projected on the screen.

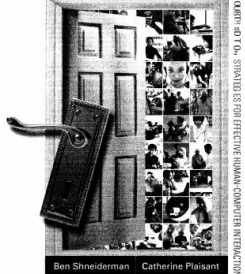
© 1987 SCIENTIFIC AMERICAN, INC.

Whole-hand Input

by
David Joel Sturman
M.S., Rutgers University (1981)
S.B., Massachusetts Institute of Technology (1979)

Submitted to the Media Arts and Sciences Section, School of Architecture and Planning,
in partial fulfillment of the requirements for the degree of
Doctor of Philosophy
at the
Massachusetts Institute of Technology
February 1992

DESIGNING THE USER INTERFACE



Ben Shneiderman Catherine Plaisant

A HAND GESTURE INTERFACE DEVICE

Thomas G. Zimmerman, Jaron Lanier,
Chuck Blanchard, Steve Bryson and Young Harvill

VPL Research, Inc.
656 Bair Island Road, Suite 304
Redwood City, CA 94063

ABSTRACT

This paper reports on the development of a hand to machine interface device that provides real-time gesture, position and orientation information. The key element is a glove and the device as a whole hndologies. Analog flex finger bending. Hand measured either by degrees of freedom, or provide six degrees of s provide the wearer of sk. These sensors are ve and connected to the ble.

and its component conjunction with a host il-time 3-dimensional the glove wearer to i objects as if they were spelling, evaluation of providing an interface ige.

piezoceramiques places sur le gant sont controles pour fournir a l'utilisateur differentes sensations tactiles. Ces differents capteurs sont montes sont un gant de faible poids, connecte au systeme de controle par un cable de faible dimension.

Les applications du systeme comprennent son utilisation relie a un ordinateur contenant un modele temps reel en trois dimensions de la main permettant a l'utilisateur de manipuler des objets virtuels generes par l'ordinateur comme s'ils existaient reellement, l'interpretation de signes alphabetiques pour mal-entendants, l'evaluation clinique de troubles fonctionnels de la main, ainsi que son utilisation comme interface a un langage de programmation visuelle.

Keywords: Human Interface, User Interface, Motor Interface, Tactile Interface, Gesture Recognition.

1. INTRODUCTION

The hand gesture input devices presented here, the Z-Glove™ and the DataGlove™, are lightweight cotton gloves containing flex sensors which measure finger bending, positioning and orientation systems, and tactile feedback vibrators.



Figure 1.
DataGlove with outer glove removed to show sensors

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David J. Sturman
Media Arts and Sciences Section
December 20, 1991

David Zeltzer
Professor of Computer Graphics
Media Arts and Sciences Section

Stephen A. Benton
Committee on Graduate Students

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Games 04
#4 of 16
Online

Virtual, Augmented and Mixed Reality

In his book *Dawn of the New Everything: Encounters with Reality and Virtual Reality* (2017), Lanier reflects on his upbringing in 1960s New Mexico, his lifelong relationship with technology, and his path to Silicon Valley. Part personal memoir and part rumination on virtual reality, Lanier highlights VR's versatility both in historical context and projects its functions into the future.

Lanier writes of VR's capacity to engage and inspire more than any other kind of technology ("TV and video games draw people into a zombielike trance ... while VR is active and makes you tired after a while"). He writes that the older, poorer VR equipment might have done an even better job at exposing one's own process of perception, since "the best enjoyment of VR includes not really being convinced. Like when you go to a magic show." And he underscores how VR inherently helps the user focus on reality, rather than the virtual world, explaining that the best magic of VR happens in the moments right after the demo ends (his lab would often present flowers to visitors coming out of the headset, as the visitor would experience them as though for the first time).

Lanier cites modern VR's rich resume beyond gaming and entertainment: it has been used to treat war veterans overcoming PTSD; by doctors to perform intricate surgeries; by paraplegics wanting to feel the sense of flight; and as a mechanism to prototype almost every vehicle fabricated in the last two decades. Throughout the book Lanier intersperses fifty-one definitions of VR, illuminating its many uses, gifts, and pitfalls. Wikipedia <https://www.youtube.com/watch?v=N0Z-KB1S-IM>

Jaron Lanier

<https://www.youtube.com/watch?v=lsNF4KfmwkY>

©

Lawnmower Man Siggraph 90's
<https://www.youtube.com/watch?v=lwbGumZ-FYg>

Online - LIVE

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Jaron Lanier VR 01
Father of Virtual Reality



Courtesy Sega



NASA Langley Research Center information technology specialist Ed McLarney demonstrates the benefits of a Microsoft HoloLens headset. (Credit: NASA/David C. Bowman)

Virtual Reality



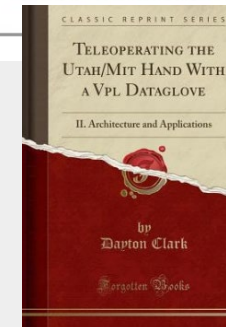
• 1985...



NASA's Virtual Workstation Shapes A VIVED Reality

Defined in Terms of Presence

- Presence is the key to defining VR in terms of experience
- Presence is defined as the sense of being in an environment
- Telepresence is defined as the experience of presence in an environment by means of a communication medium.
- A "virtual reality" is defined as a real or simulated environment in which a perceiver experiences telepresence.



Source: NASA

©

Hololens, Magic Leap, Oculus, Varjo

Online - LIVE

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Virtual Reality today has the tools
the horsepower, the bandwidth
necessary to be useful on any device

In this photograph from June 7 1989, two people demonstrate the EyePhone system which uses special goggles and a DataGlove[®] which allows them to see and move objects around in a computer created environment. The EyePhone, developed by VPL Research, was on display at the Texpo Telecommunications Show held in San Francisco

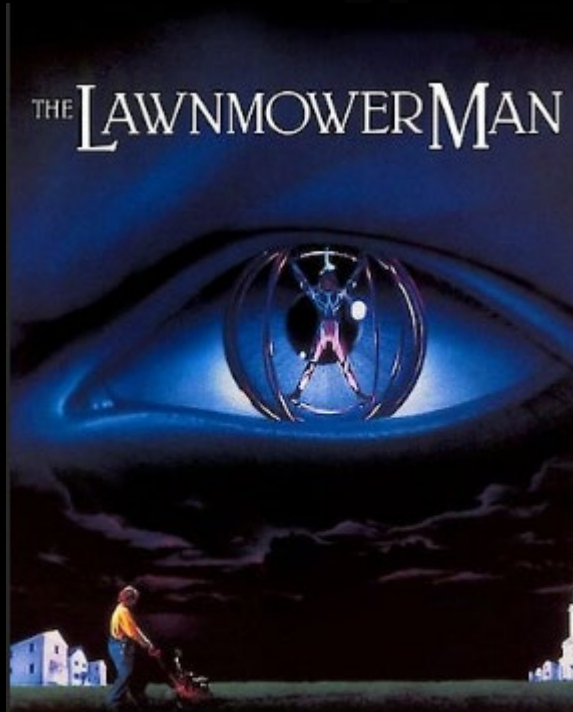
Jaron Lanier's EyePhone and DataGlove 1988



VPL Research, filed for bankruptcy in 1990 and in 1999, all of its patents were bought by Sun Microsystems

©

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1992 Film



[The Lawnmower Man \(film\) - Wikipedia](#)

[The Lawnmower Man \(1992\) - IMDb](#)

©

Siggraph, French animation & DEC Alpha
helped move early V.R. forward

26 of 82

Virtual Reality was considered the
arrival of Sci-Fi capabilities

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G Theory 04
#4 of 16
Online



MACE Virtual Labs and Houston Community College teamed up to create a new virtual reality lab at the HCC West Loop campus. *Photo via hccs.edu*

<https://www.hccs.edu/about-hcc/news/articles/hcc-in-rare-air-with-opening-of-new-virtual-reality-lab.html>

©

Suits, Goggles, Cars, Treadmills, Studios

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Houston Community College's
Stellar Virtual Reality Lab

Online - LIVE

Augmented and Mixed Reality

Virtual Reality (VR) has been the “next big thing” for several years, but its time has finally come as a way to generate realistic images, sounds, and other sensations that put you smack in the middle of a spectacular imaginary world. Augmented Reality (AR), which adds virtual stuff to your real-world environment, is contributing to the buzz, and both technologies should become a big part of our future. With Mixed Reality (MR), you can play a virtual video game, grab your real-world water bottle, and smack an imaginary character from the game with the bottle. Imagination and reality have never been so intermingled.

So much is happening so fast that the differences between VR, AR, and MR can seem a little puzzling at first. Each of these spellbinding technologies are accessible to almost everyone, but before you throw down your hard-earned money for the latest head-mounted display, let’s take a closer look at what you’ll need for an amazing VR, AR, or MR experience. Source: Intel

©

Mixed Reality is the mixing of different environments to gain from all abilities



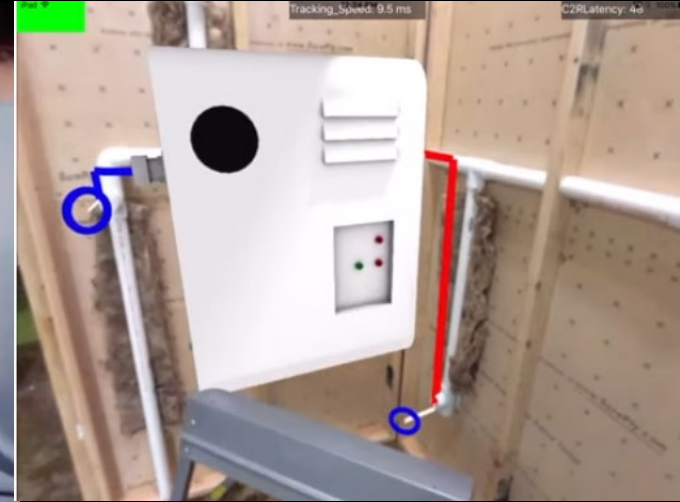
Augmented Reality is the placement of artificial elements in a real environment

Augmented and Mixed Reality

Courtesy Occipital



Measure your rooms automatically



Place objects in the worksite

The Occipital Structured Sensor fits right onto an IPAD and can measure and place any type of object in real-time

©

<https://www.youtube.com/watch?v=BPgIKtqygL8>



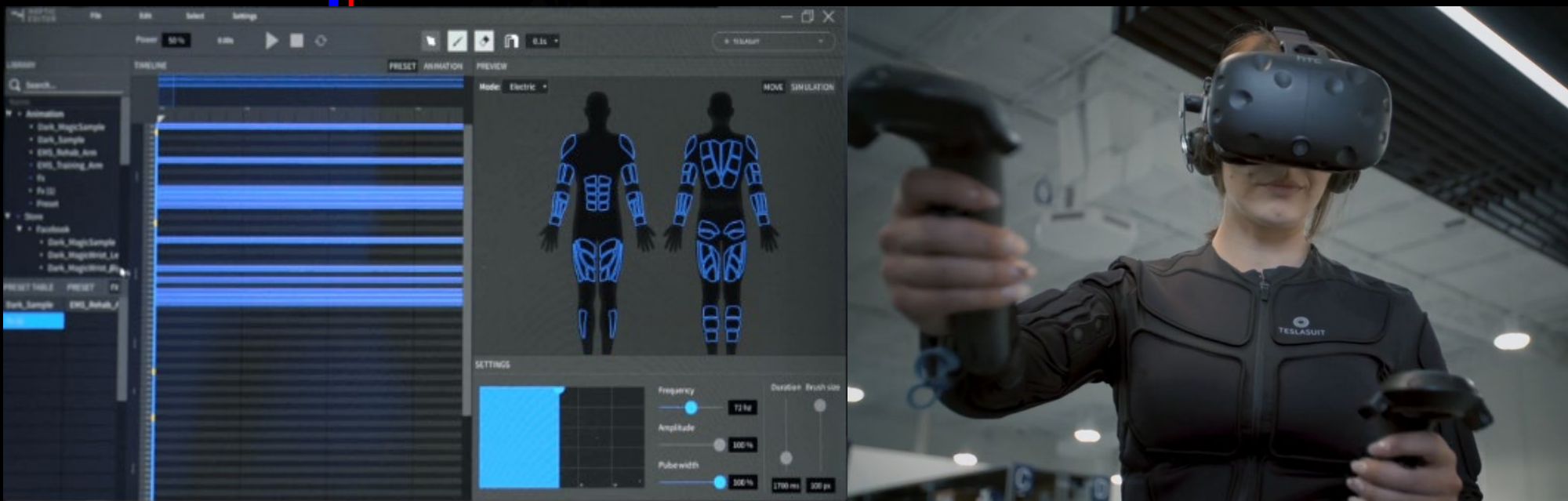
Augmented Reality is the placement of artificial elements in a real environment

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G Theory 04
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Online

Maximizing Haptic Performance



Courtesy TeslaSuit

<https://www.youtube.com/watch?v=OEMa2-IVfyw>

©

Goggles, Gloves, Shoes, hats & full suit

Online - LIVE

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The new Tesla Suit is full reality haptic
feedback in any situation
complete with gloves, shoes, hats, etc.

Courtesy Varjo



The driver can see the real world as well as “placed unreal items” that appear real

©

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Online

AMC Halt and Catch Fire Season 2



Courtesy AMC

<https://www.youtube.com/watch?v=o9E28XNa6ko>

©

Interesting Series Video Games Start



Historical Series about Video Games

Online - LIVE

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Games
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Online



Historical Topics



SCORE 0000
TIME 01 07
FUEL 0692

ALTITUDE 139
HORIZONTAL SPEED 5
VERTICAL SPEED 83



MoonLander, 3D tic-tac-toe and Lunar Lander

©

Arcade games moved to PC's Space Invaders



Early days of Computer Game Engines

Online - LIVE

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A.I. in Scoring is the Realtime Assessment Tool



Atari's classic shooter, released in 1979, **was** the first **game** to allow **players** to create a personalized **top** 10 table for that day's **scores**. GWR Video Gamer's Edition 2008, title: First video **game** to include a personalized **high-score** table.

©

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Online

A.I. in Today's Games



Courtesy Gaijin Entertainment



<https://www.youtube.com/watch?v=faKyQOTmzDw>

Online - LIVE

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Realistic WarThunder Battle Games 01
Off the shelf great realism

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Online

Game Laboratories

ASU Center for
Games & Impact
Arizona State University



USC



<https://game.engineering.nyu.edu/people/#>

NYU Game Innovation Lab

Online - **LIVE**



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<http://www.american.edu/gamelab/>

USC: Keynote with Tracy Fullerton - Connected Learning Summit 2019

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Games 02
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Online

Hot Topics Today

Game Engines



Courtesy Black Panther Animated

©

You can literally make your own components



Computer Game Engines are very powerful and customize-able

Online - LIVE

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G Theory 04
#4 of 16
Online

How to create a configurable car in Unity HDRP Edward Martin, January 23, 2019

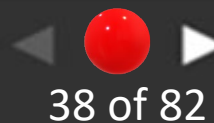


Imagine how many things you could begin to configure

©

<https://vimeo.com/294134490>

Online - LIVE

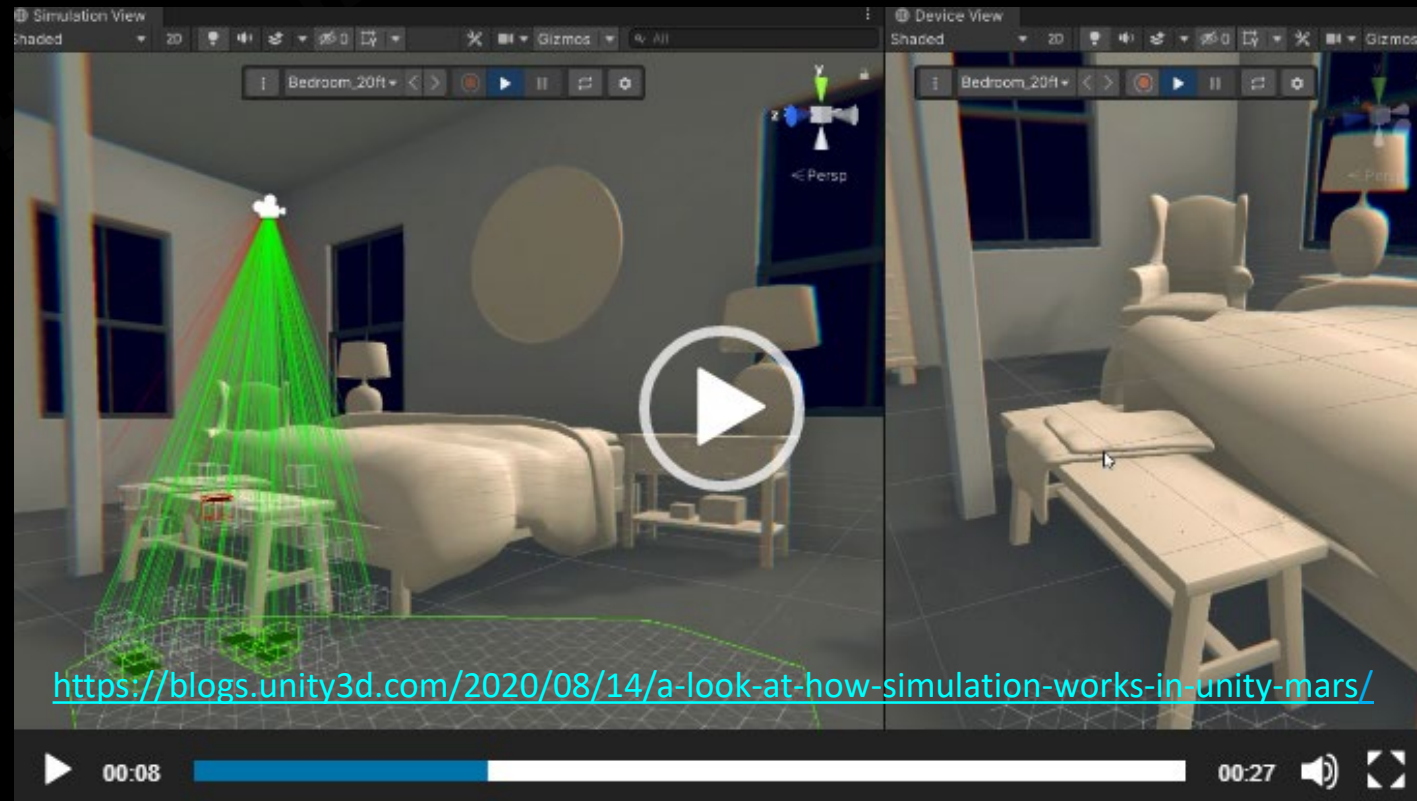


This is a good example of being
able to "Roll your Own"

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Online



<https://blogs.unity3d.com/2020/08/14/a-look-at-how-simulation-works-in-unity-mars/>

Real Estate, Architecture, Design, Construction

©

<https://www.superoffice.com/blog/saas/>

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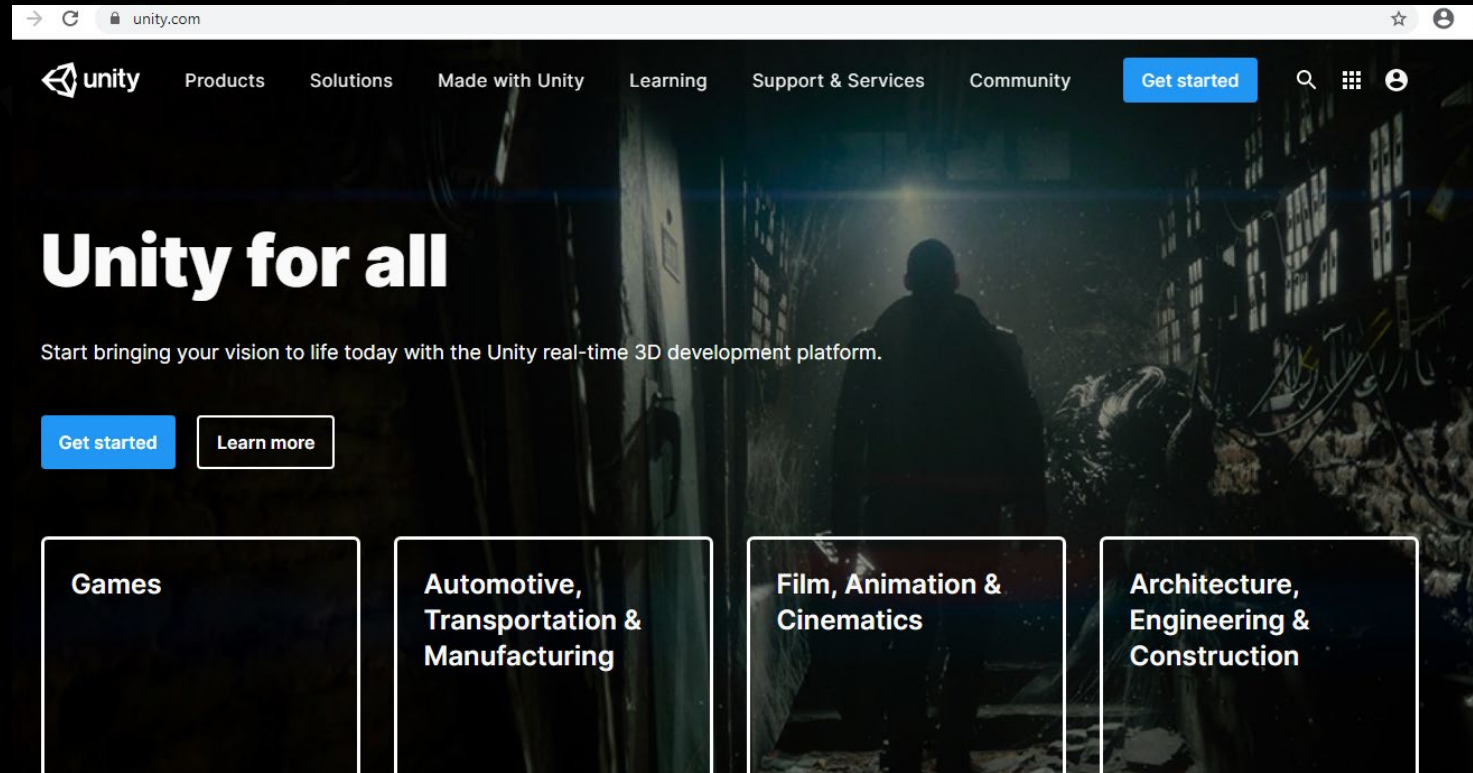
Unity MARS is a simulated environment
where everything is changeable

Online - LIVE

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G Theory 04
#4 of 16
Online

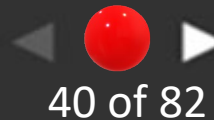


Courtesy UNITY

It used to take years to produce a game

©

Real-time Simulation along side
Real-Life is a winning combination



Gaming is now simulation, and it is
being utilized almost everywhere

A.I. 1370



G Theory 04
#4 of 16
Online

Courtesy UNITY



https://www.youtube.com/watch?time_continue=5&v=4KutQxCya0g&feature=emb_logo

©

Game Engines provide all services

Online - LIVE



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You can make a game, change a game,
publish a game and distribute a game
with the same game engine

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G Theory 02
#2 of 16
Online

Is it Real, a Game or a Simulation?



Courtesy Octane Render

[OTOY • News](#)

[OTOY • OctaneRender: Showcase](#)

©

Adding A.I. gives you more control

Online - **LIVE**

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Simulate anything that would help
you understand the real world, work
or cultures, societies and people

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G Theory 02
#2 of 16
Online

RNDR as a SaaS

Stunning Business Model of The Future



Courtesy Octane Render



<https://rendertoken.com/>

©

<https://home.otoy.com/octane2020-rndr-released/>

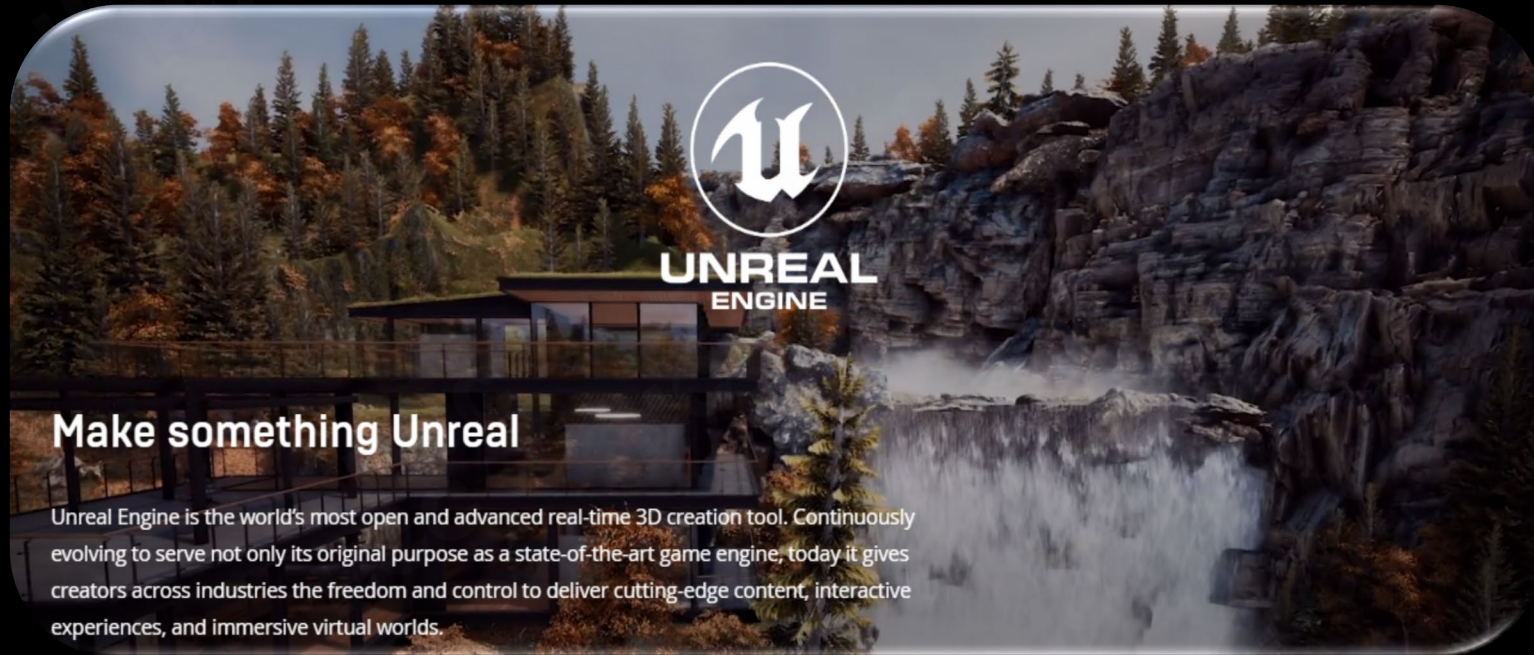
Adding A.I. gives you more control



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Simulate anything that would help
you understand the real world, work
or cultures, societies and people

Photorealistic Properties of Games Drive Physics



Courtesy Unreal Engine

Life-like Properties with behaviors are fed by precision Physics Engines

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G Theory 02
#2 of 16
Online



VRWorks enables a new level of presence by bringing physically realistic visuals, sound, touch interactions, and simulated environments to virtual reality.

©

Nvidia knows how to channel the horsepower
[NVIDIA VR Platforms Deliver Massive Performance Boost for Virtual Reality | NVIDIA Newsroom](#)

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Virtual Reality, AR and MR provide cutting edge tools to visualize Big Data



"You have brains in your head
and feet in your shoes,
you can steer yourself in any
direction you choose!"

--Dr. Seuss

In 2D, 3D and 4D

©

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G Theory 02
#2 of 16
Online

Augmented or Mixed Reality



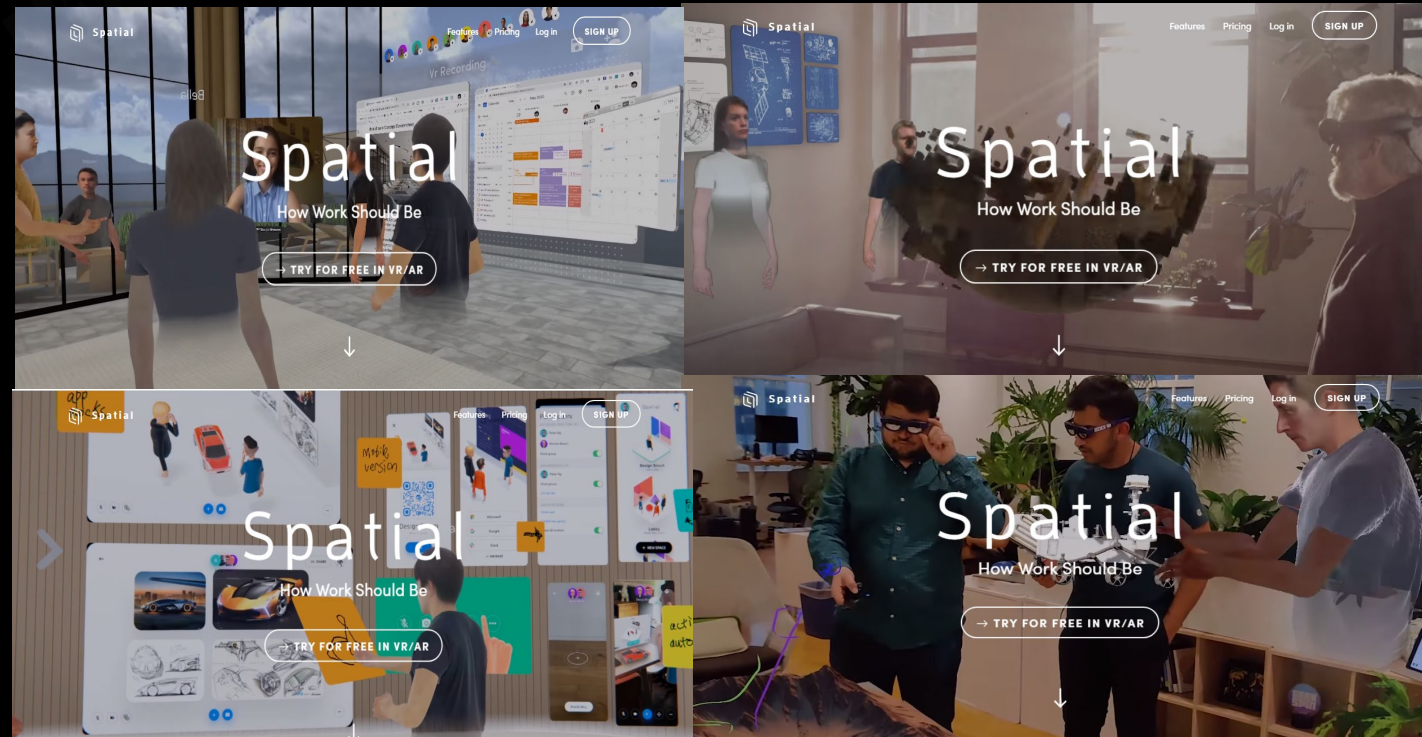
[Spatial - Your World Awaits](#)

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Augmented Reality 01

"Spatial is a collaborative, holographic, augmented reality solution,"

Augmented or Mixed Reality



Different workstyle Environments

©

How Close Are We To The HOLODECK?



<https://www.youtube.com/watch?v=bOf2S7OzFEg>

NVIDIA HOLODECK launches new VR Tools

©

Perhaps we have surpassed the holodeck
Online - **LIVE**

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Jensen Huang's Keynote 2020
Nvidia – if you want to hear about
the future happening now.

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G Theory 02
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Online

NVIDIA HOLODECK launches new VR Tools for
Engineering, Architecture, Manufacturing and Construction

By Rebecca Hills-Duty — Last updated Aug 9, 2018



Audi: https://www.youtube.com/watch?v=6Z4n6KzUgPc&feature=emb_rel_end

©

<https://www.youtube.com/watch?v=KcWZFpjjD34>



Online - LIVE

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Nvidia's Holodeck and then see
how Audi is making use of it

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G Theory 02
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Online

Platforms



Steam is one of the largest Gaming and VR Platforms Available
Supports most all hardware and content

©

Secure, flexible and very fast
Online - **LIVE**

◀  ▶
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In essence, VR is Mobile and
highly distributed

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G Theory 02
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Online

You may not have a portal, but you can
at least experiment on one



[Portal on Steam \(steampowered.com\)](https://store.steampowered.com/app/431960/Portal)

©

Online - LIVE



STEAM™ BOX

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Portal is a new type of game that
simulates a scientific laboratory

Portal™ is a new single player game from Valve. Set in the mysterious Aperture Science Laboratories, Portal has been called one of the most innovative new games on the horizon and will offer gamers hours of unique gameplay.

The game is designed to change the way players approach, manipulate, and surmise the possibilities in a given environment; similar to how Half-Life® 2's Gravity Gun innovated new ways to leverage an object in any given situation.

Players must solve physical puzzles and challenges by opening portals to maneuvering objects, and themselves, through space.

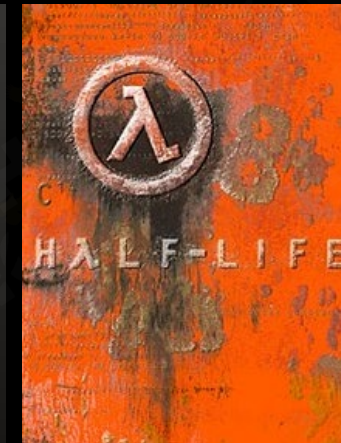
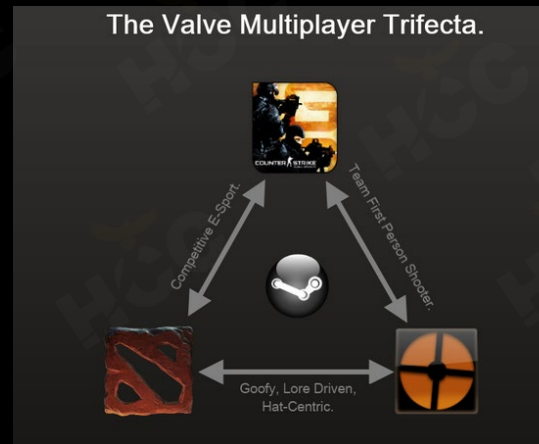
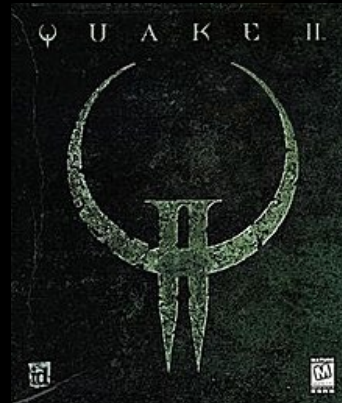


How do we get acquainted with technology we don't have yet?



Valve Corporation, also known as Valve Software or simply Valve, is an American video game developer, publisher, and digital distribution company headquartered in Bellevue, Washington. It is the developer of the software distribution platform Steam and the Half-Life, Counter-Strike, Portal, Day of Defeat, Team Fortress, Left 4 Dead, and DOTA series.

Valve was founded in 1996 by former Microsoft employees Gabe Newell and Mike Harrington. Their debut product, the PC first-person shooter Half-Life, was released in 1998 to critical acclaim and commercial success, after which Harrington left the company. In 2003, Valve launched Steam, which accounted for around half of digital PC game sales by 2011. By 2012, Valve employed around 250 people and was reportedly worth over US\$3 billion, making it the most profitable company per employee in the United States. In the 2010s, Valve began developing hardware, such as the Steam Machine, a brand of gaming PCs, as well as the HTC Vive and Valve Index virtual reality headsets.



©

Digital Distribution is a Critical Subject

Valve is one of the most successful game companies on the planet. It helped usher in the idea of digital distribution. The company's version of an online app store, Steam, is known for selling games made by third parties, such as Skyrim and Call of Duty. But Steam's real value lies in the way it gives independent publishers equal footing against big corporate game makers.

What the numbers show is that Valve has been incredibly successful at converting its in-house games into hits. Steam users have spent a combined 3 billion hours playing Valve's fantasy multiplayer game, Dota 2, and nearly 1.5 billion hours playing the cartoon-y combat simulator Team Fortress 2. The overall gaming universe is huge, and Valve is just a part of it; you can't play Steam games on consoles, such as the Xbox, for instance. But at least in the space it controls, Valve has somehow managed to outmaneuver even the most powerful titans of the industry.

The more highly distributed a game like DOTA2 is the more data for A.I. It is easy to see why massively online, multi-agent interactivity is essential.

[Dota 2](#)

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G Theory 02
#2 of 16
Online

Pipelines

VALVE

We make games, Steam, and hardware. Join us.

We're seeking collaborators who are the best at what they do. Opportunities at Valve run broad and deep across the following areas, where we're always hiring for all positions.

Art

Dota 2	3,828
Team Fortress 2	1,444
Counter-Strike: Source	1,281
Counter-Strike	1,251
Sid Meier's Civilization V	743
The Elder Scrolls V: Skyrim	632
CoD: MW2 - Multiplayer	625
Garry's Mod	385
Counter-Strike: GO	362
Left 4 Dead 2	297
CoD: MW3 - Multiplayer	263
Empire: Total War	249
Terraria	218
CoD: Black Ops - Multiplayer	207
Borderlands 2	185
Football Manager 2014	143
Total War: SHOGUN 2	137
Fallout: New Vegas	137
Counter-Strike: Cond. Zero	114
Left 4 Dead	106

ars

Beautiful Platform for A.I.

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Online - LIVE



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Games, programming, Steam,
Hardware, distribution & community
all matter especially with A.I.

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G Theory 02
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Online

Internationally

Simulation

Footage



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International Simulation is showing
Real-time Sim-Ops with Live Realtime

Games Now Mean Simulation

Automotive,
Transportation &
Manufacturing

Gaming

Film, Animation &
Cinematics



Architecture,
Engineering &
Construction

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You Can Be Effective Everywhere



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Think and Work Differently, more effectively

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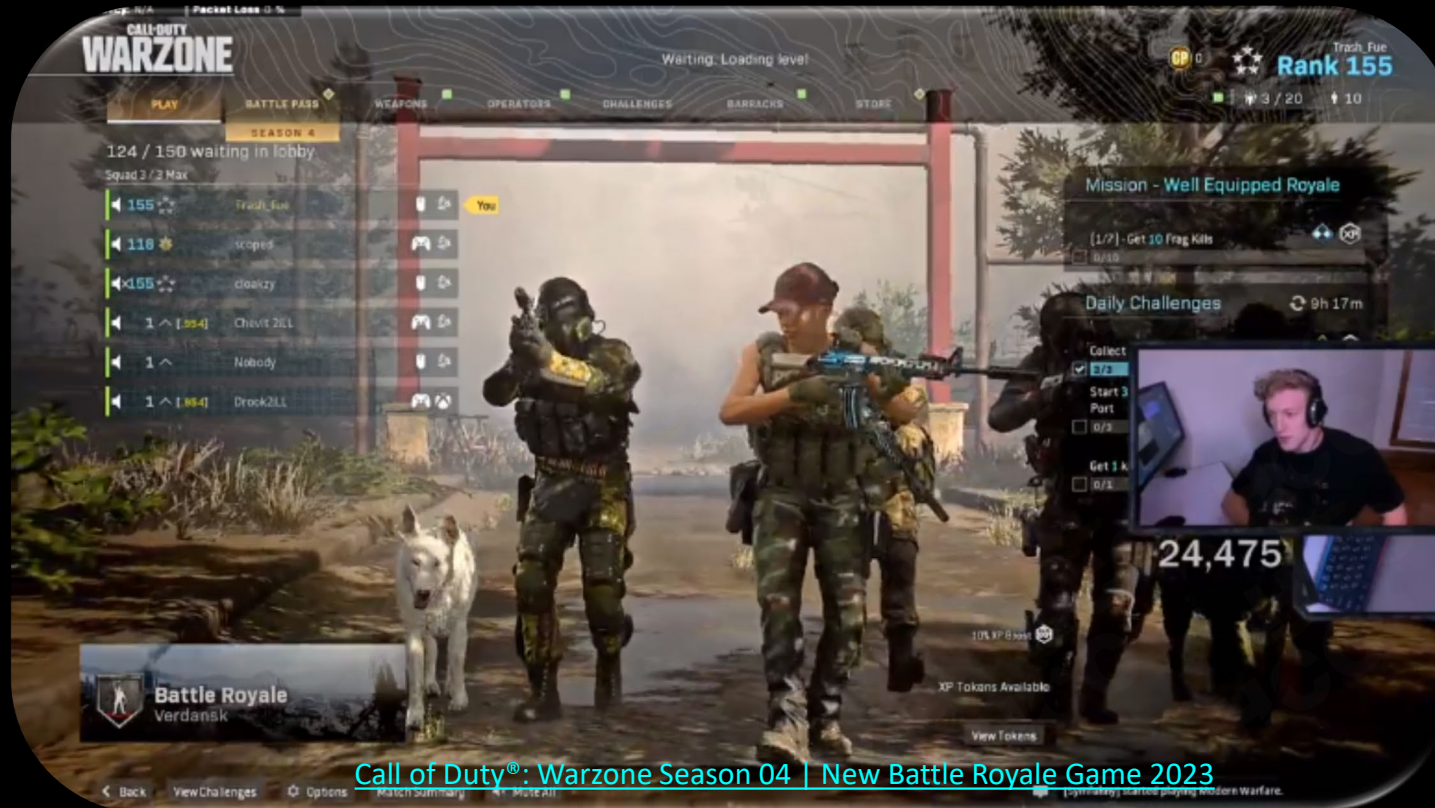
Hololens, Unity, A.I., lots of great
Data, Machine Learning & GPU's

Online - LIVE

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Games 04
#4 of 16
Online



[Call of Duty®: Warzone Season 04 | New Battle Royale Game 2023](#)

Voice, Video, Interactivity, Collaboration, Real Time

©

Safe Experimentation
Online - LIVE

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Realtime-ish Play and Real-time
Scenarios and interactive collab.

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Games 04
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Online



The **ESPN+** of Game Play

By 2015, Twitch had more than 1.5 million broadcasters and 100 million viewers per month.^[6] As of 2017, Twitch remained the leading live streaming video service for video games in the US, and had an advantage over YouTube Gaming.^{[7][8]} As of May 2018, it had 2.2 million broadcasters monthly and 15 million daily active users, with around a million average concurrent users.^{[9][10]} As of May 2018, Twitch had over 27,000 partner channels.^{[9][11]} Wikipedia

©

LIVE action Feeds [Twitch](#)



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Broadcasting activity is essential
and another dimension for A.I.

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Games 04
#4 of 16
Online

Live Online, Really

©

Online - **LIVE**



Work, shop, Play, Interact ONLINE

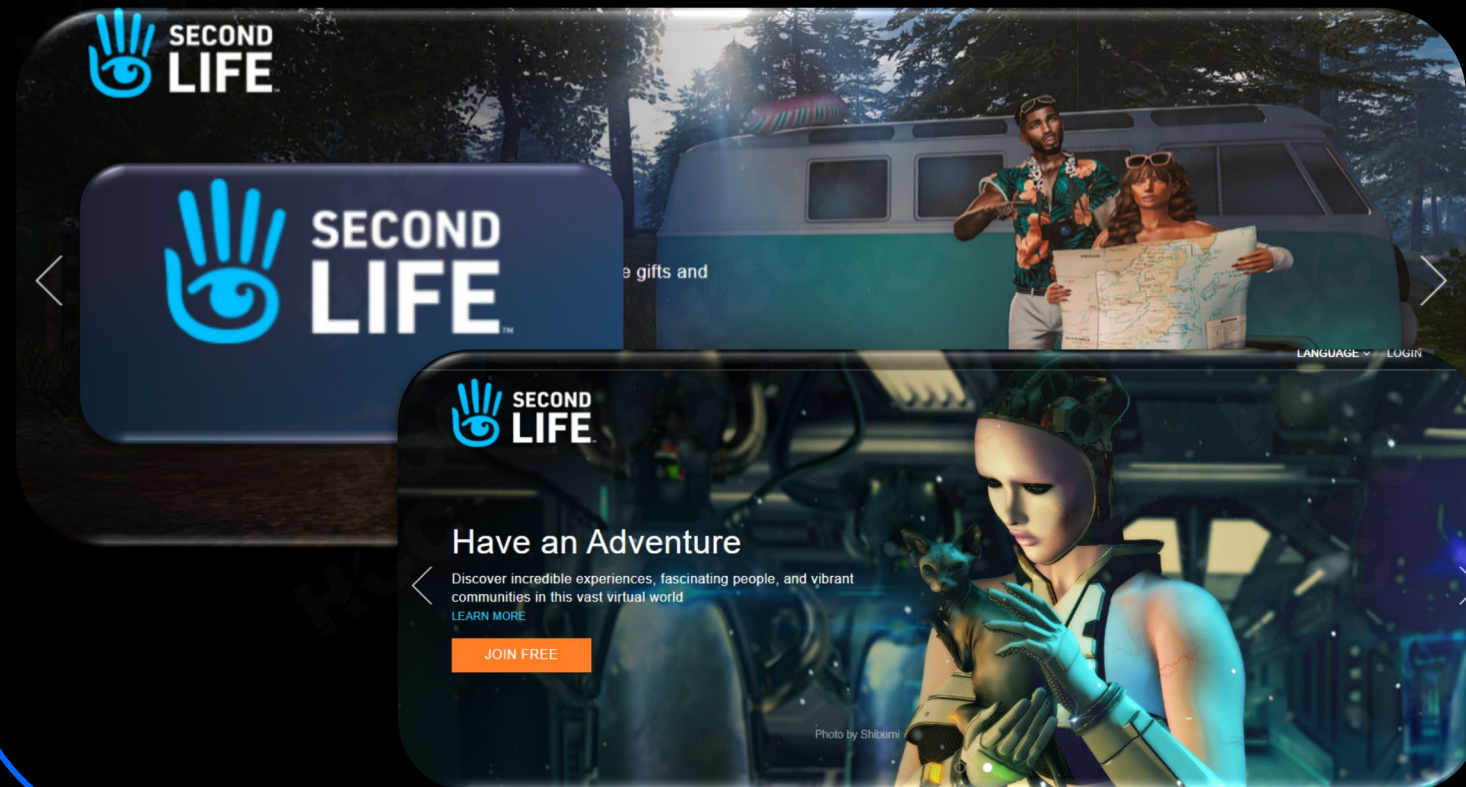
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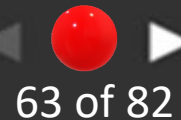
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#2 of 16
Online

Virtual Communities



©

First to offer Tech conferences online



Second Life Communities have
Stores, businesses, schools, etc.

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G Theory 02
#2 of 16
Online



'Second Life' still has dedicated users in 2020. ... To put it briefly, **Second Life** is a virtual, online world where users create avatars which can travel to worlds and lands (called Sims), participate in role-playing games, create and sell products, and socialize with other **Second Life** Residents. Feb 16, 2020

<https://secondlife.com/>

Online - **LIVE**

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Try out Second Life – it has been around for a long time

A.I. 1370



Realtime
#2 of 16
Online

“Sim Real Life in Realtime” Right Now

SELF-DRIVING CARS PRODUCTS SOLUTIONS PARTNERS FOR DEVELOPERS

Products

https://www.youtube.com/watch?v=OT_MxopvfQ0

NVIDIA DRIVE
CONSTELLATION

Virtual Reality Autonomous Vehicle Simulator

<https://www.youtube.com/watch?v=iTZ9GoN2Q0k>



<https://www.youtube.com/watch?v=aQwqD5cB2ck>

©

<https://www.youtube.com/watch?v=ftsUg5VlzIE>

Surround Cameras make a huge difference

Online - LIVE

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Visit Nvidia's Drive Lab and take
A look at their Drive Constellation

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Octane
#2 of 16
Online

The Future of GPU Rendering RaaS

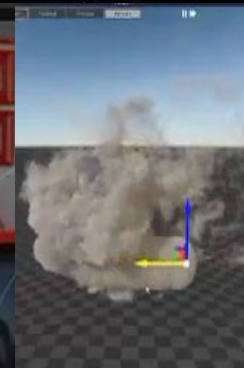
CAPTAIN
MARVEL



Out of Beta



Octane X on iPhone as a Render Farm



EmberGenFX

©

RTX | 'Always on'
Online - LIVE



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Otoy, Inc. RNDR Token on Blockchain
RaaS GPU Network



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Otoy
#2 of 16
Online

The Future of GPU Rendering



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Changing Engineering



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Otoy, Inc.

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Otoy
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The Future of GPU Rendering



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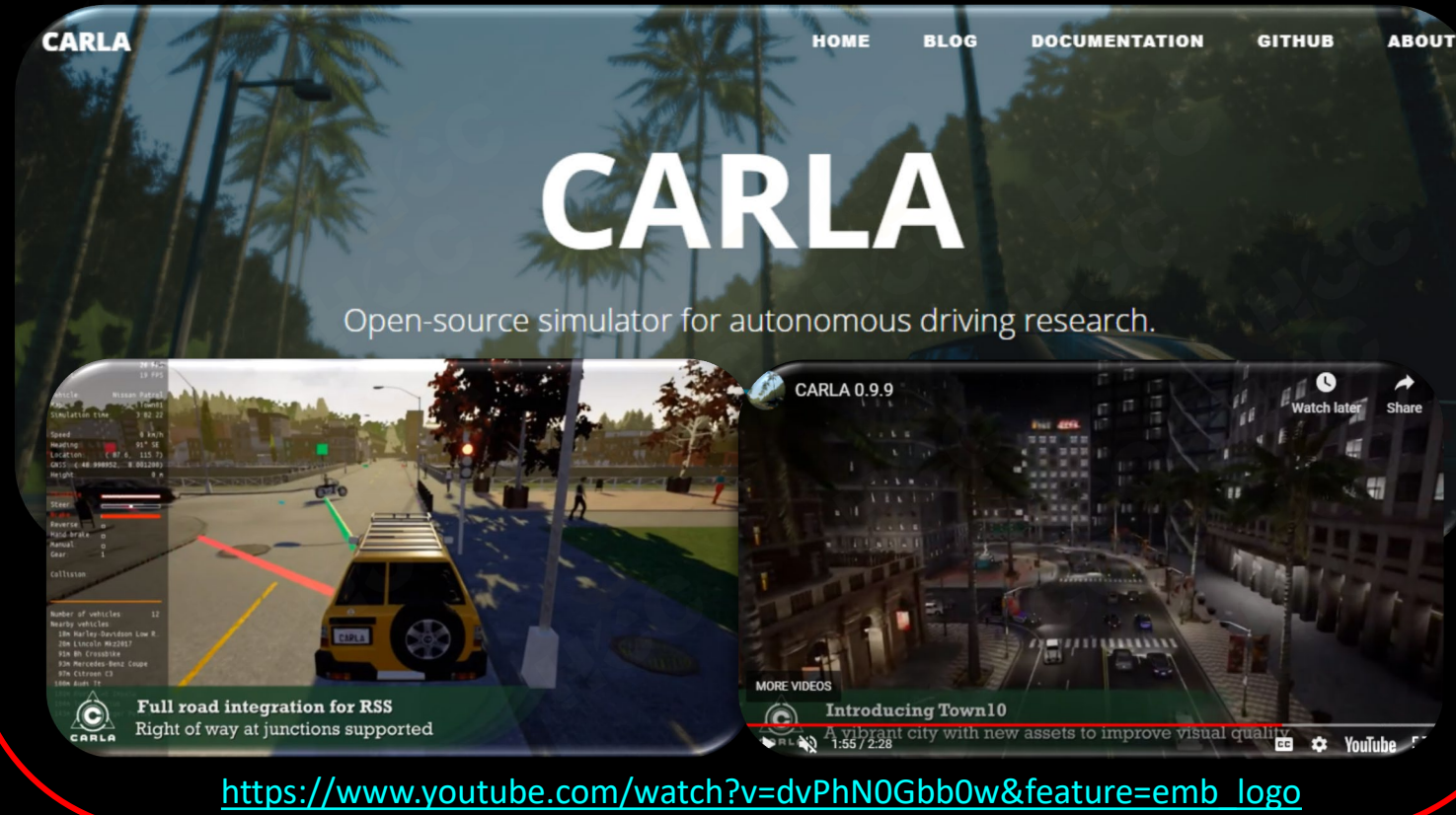
Nvidia's new Headquarters
Otoy, Inc.

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G Theory 02
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Online

Autonomous Driving Simulator



<https://carla.org/>
Online - LIVE

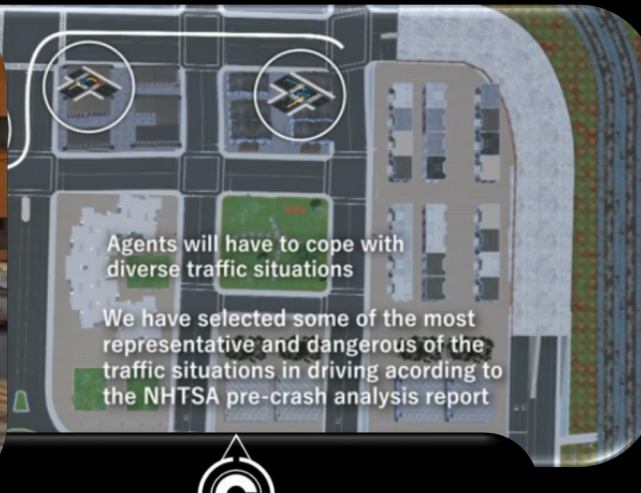
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Autonomous Driving Simulator 01
Open Source, Multi-agent, RL

The Perfect Practice Environment



Introducing Town10
Intricate buildings and streets improve visual quality



Agents will have to cope with diverse traffic situations

We have selected some of the most representative and dangerous of the traffic situations in driving according to the NHTSA pre-crash analysis report



CARLA

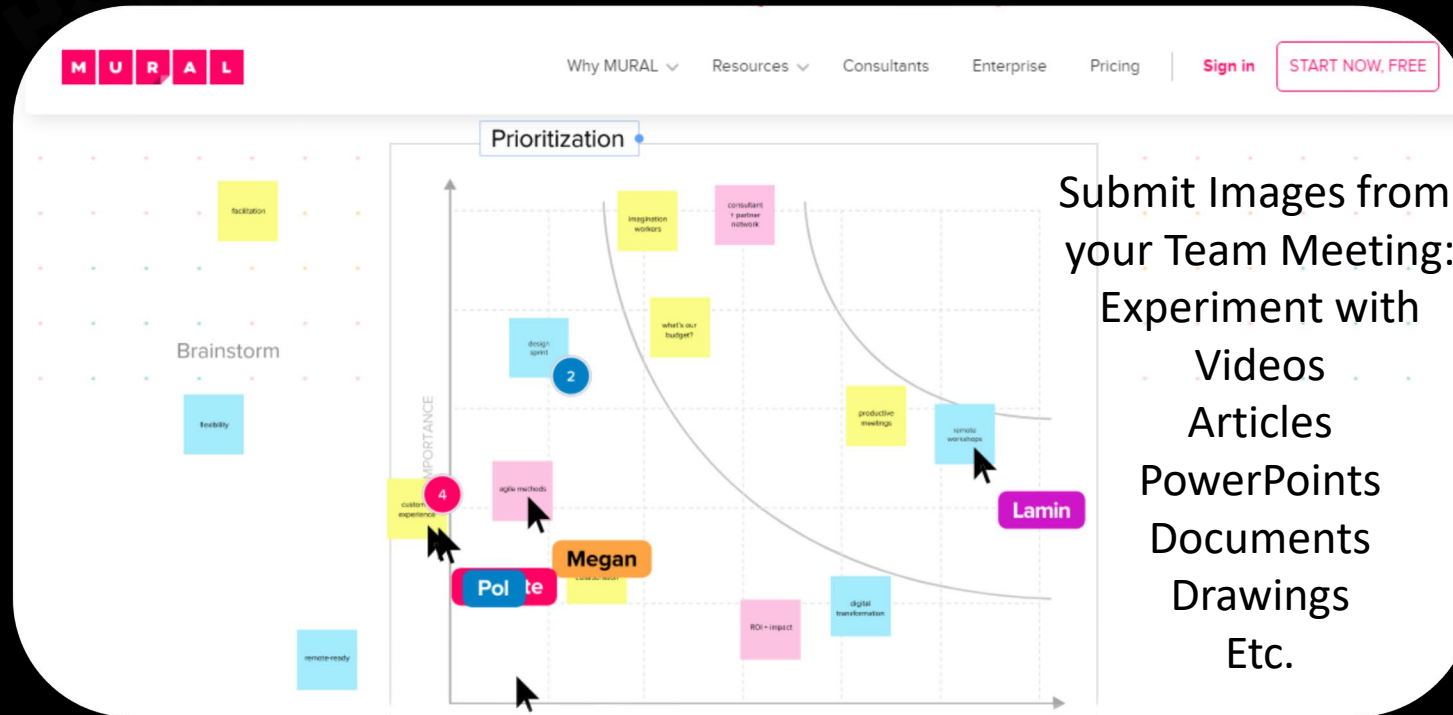
Hosted by:



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Laboratory 04



Get used to Working this way and then imagine how this would work in Spatial

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Lecture
Notes 03

MAKE ME LAUGH CHALLENGE

PART 9

Have a great joke ready for your team. we will go thru what jokes are about.

©

Each Team needs a joke each week

Online - **LIVE**



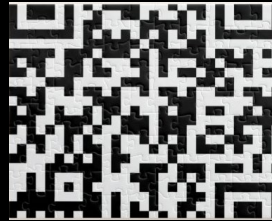
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Be prepared to tell a joke
or a funny story, riddle, or toast

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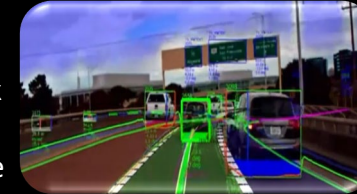


Lecture
Notes 02



QR Code of the Day
Augmented Reality Code

Big Idea of the Week
Nvidia Drive Labs
Predicting the Future

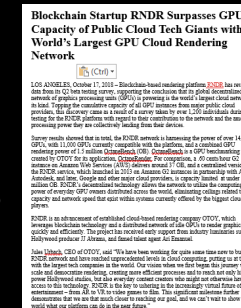


Key Video
of the Week
Baby X



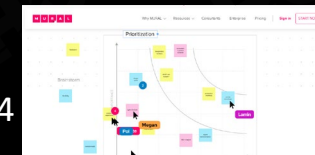
Leadership Skill 04
Learning to Learn

Key Article
of The Week
RNDR as SaaS



What is Raytracing?
Homework

Join Mural with your Team
Use Different Media Lab L04



Weekly GEMS that help Learning
Online - LIVE

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A.I. GEMS for The Week with
a brief review of last week

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Lecture
Notes 02

Leadership Skills Opportunity 04



Barbara Oakley
Learning How to Learn

February 10th, 2015

<https://www.youtube.com/watch?v=vd2dtkMINIw>

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Yet another great episode in Business Skills
Online - LIVE



Leadership skills 04-Learn to Learn

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Assignment 04

What is Raytracing?

Prepared with a 5-minute presentation about the subject

This may be one of the most important technologies of our time
If you maximize its capability with A.I.

<https://www.youtube.com/watch?v=gBPNO6ruevk>

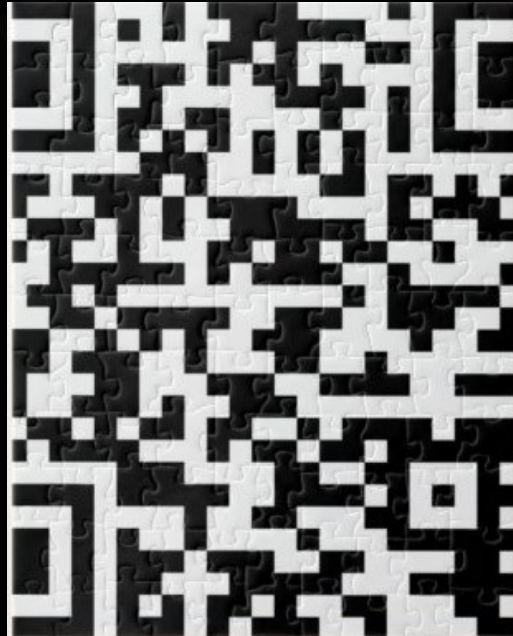
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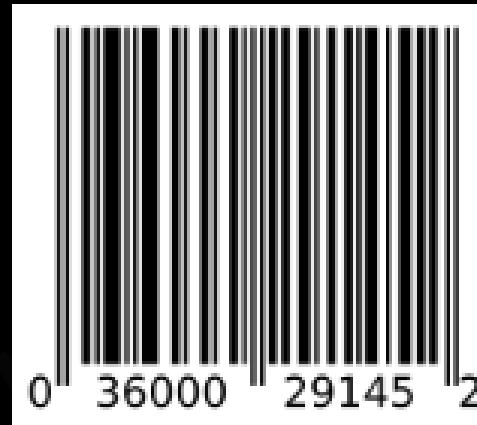
G Theory 04
#4 of 16
Online

Puzzle of The Day?



QR Code

What is it?
What does it do?



Barcode

What is it?
What does it do?



What type
of codes are
being used in
Augmented Reality

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RNDR Network User Manual



Section I: General

- A) About RNDR
- B) Architecture & Network flow
- C) Network Roles
- D) Network Core Technologies
- E) RNDR Compute Pricing
- F) RNDR Tokens and Currency Gateways
- G) Governance and Incentives
- H) Security
- I) Watermarking
- J) Reputation Score
- K) Job Assignment
- L) Job Approval and Dispute Resolution
- M) Proof-of RNDR
- N) Asset Hashing
- O) Transaction Fees
- P) Network attack vectors & mitigation
- Glossary

RNDR User Manual 2020

1

The RNDR Manual covers everything you might like to know

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G Theory 04
#4 of 16
Online

otoy Shop Downloads Capture Render News Forum Support About SIGN-IN

LightStage

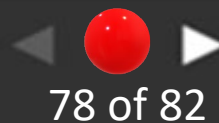
A cutting-edge scanning service for VFX and Games.

→ WATCH LIGHTSTAGE IN ACTION

<https://home.otoy.com/capture/lightstage/>

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Industrial light and Magic utilizes commercial grade facial mapping



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Dynamic Facial expression linked to Speech is Deep Learning Driven

Academy Award® Winning Technology

LightStage™ is a cutting-edge scanning technology that allows for the digital capture of the human face with unparalleled fidelity and scientific accuracy. LightStage efficiently captures how a subject's face appears when lit from every possible lighting direction. From this captured imagery, specialized algorithms create realistic virtual renditions of the subject in the illumination of any location or set, faithfully reproducing the color, texture, shine, shading, and translucency of the subject's skin.

For over a decade, LightStage scans have been the go-to for blockbuster films and AAA game titles alike. Co-created by LightStage CTO Tim Hawkins, the technology behind LightStage is based on original research at the USC Institute for Creative Technologies. In 2008, OTOY licensed the technology for commercial use and opened the LightStage capture facility in Burbank, CA. LightStage won a Scientific and Engineering Award from the Academy of Motion Picture Arts and Sciences, that [cited](#) its "ability to capture high fidelity reflectance data of human subjects, allow[ing] for the creation of photorealistic digital faces as they would appear in any lighting condition."



Everything from models, news anchors, presenters, teachers, actors and avatars

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G Theory 04
#4 of 16
Online



Mark Sagar started his career by building medical simulations of body parts. He took those skills and went into CGI, most famously for movies including Avatar, King Kong, and others. Now he's combining his skills and building an entire brain and responsive face on a computer in order to make a relationship seem real.

https://www.youtube.com/watch?time_continue=16&v=yzFW4-dvFDA&feature=emb_logo

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Baby X is learning to play Pong

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Online - LIVE

BABY X: DIGITAL, ARTIFICIAL, INTELLIGENCE, COMPUTATIONAL NEUROSCIENCE AND EMPATHETIC INTERACTION

Deborah Lawlor-Dorman, Doctoral Candidate,
National Institute of Creative Arts and
Industries, University of Auckland,
Auckland, New Zealand and College of
Fine Arts, University of New South Wales,
Sydney, Australia, [1]
E-mail: dlaw111@unsw.edu.au

Submitted: cleave for Editor to date

Abstract.

As a new media curator, I work with
artistic practices that engage multi-
sensory media environments. *Baby X* is a
digital artificial intelligence mixed reality
installation created by Dr Mark Sagar.
It is a contemporary neuro-behavioural
computational model with emergent
behaviours actively being used for neuro-
scientific research and, at times, a
media art installation on public display.
This paper will explore some of the di-
verse issues at play in this project from
the perspectives of embodied cognition,
emotional engagement and perception
within a mixed reality environment and
trans-disciplinary research context.

Key Words: media art installation,
mixed reality, embodied cognition



Fig. 1. Sagar, M with Laboratory for Artificial
Technologies, Auckland Biomechanics Insti-
tute, Baby Y (simplified, autonomous animation,
2013)

In the Laboratory for Artificial Tech-
nologies at the Auckland Biomechanics
Institute at University of Auckland, Dr
Mark Sagar and his team are building a
computational model of the brain and
face. The model is constructed using
current neuro-scientific research sourced
through collaboration with the universi-
ty's Centre for Brain Research.

Baby X is an aspect of this research
project. Having come from the animation
film and entertainment industries,
Sagar's research interests include devel-

oping a universal system for mapping
faces including morphology, anatomy
and biomechanical modeling. Accord-
ingly he analyzed, the muscular facial
control system and began investigating
the emotions and neuro-behaviours driv-
ing character expression. This led, in
turn, to the current project to create a
neuro-behavioural computational model
with emergent behaviours.

In doing so, Sagar has developed a
unique aspect of this research project: a
mixed reality installation for display in
public settings including museums and
galleries. When installed the work, *Baby X*,
is essentially an interactive media art
installation, containing both scientific
and artistic content. *Baby X* involves a
number of challenges to the new media
curator in terms of contextualization,
presentation and audience engagement.
The work traverses a diverse disciplinary
territory, it has a complex technical na-
ture, it raises specific debates concerning
embodied cognition and neuro-
behaviours, it is displayed in both scien-
tific and artistic venues and has a wide
and deep collective and specialist re-
search culture supporting its ongoing
evolution.

In a typical 'media art' installation
Baby X exists within a standard black
box gallery environment. Upon entering
the exhibition, the audience sees a large
screen on which the face of an animated
child is projected, disembodied, larger
than life and floating against a black
background. This autonomous animation
is capable of responsive behaviours
through a variety of camera-based sensor
tracking systems aimed at monitoring
eye and individual body movement of
the viewer. The child follows their ac-
tions and movements, often engaging
through direct gaze, and displaying re-
sponsive behaviours such as smiling, crying,
confusion and abandonment. The child's
character is based on Sagar's own six-
month-old baby and is therefore pre-
sented.



Fig. 2. Sagar, M with Laboratory for Artificial
Technologies, Auckland Biomechanics Insti-
tute, Baby Y (animation with autonomous ani-
mation, 2013)

Linked to the autonomous child ani-
mation, separately rendered, is a real-
time neural simulation. In a live neural
network, representations of muscular
anatomy through to the neuronal activity
and neuromodulator levels can be
viewed. It is possible to follow live stim-
uli through the responsive neural path-
ways and through this simulation gain
understanding of neural networks. Thus,
Baby X brings into play an oscillation
between character, neurobiological rep-
resentation and digital artificial intelli-
gence. Digital artificial intelligence in
this sense is a simulated virtual biology
based on life-like behaviours. Digital
artificial intelligence in the case of *Baby X*
can be encapsulated, as Shanon has
described, as "synthetic biology". [2]
In this sense, *Baby X* digitally simu-
lates a synthetic biology through repre-
senting neurobiology generated from
algorithmic functions. This project is
made possible by its trans-disciplinary
research basis. *Baby X*, as a work, cross-
es across disciplinary boundaries, being
of interest to the arts, sciences and bio-
engineering. Mark Sagar as a practitioner
is variously an animator, designer and
bio-engineer.

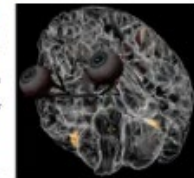


Fig. 3. Sagar, M with Laboratory for Artificial
Technologies, Auckland Biomechanics Insti-
tute, Baby Y (animation with autonomous ani-
mation, 2013)

This trans-disciplinary is reflected in
its design framework. *Baby X* is con-
structed of "neural building blocks". It is
built to be both modular and to accom-
modate the needs and interests of diverse
research communities. As Sagar states:
"We are building a collaborative modu-
lar model of the face and brain, a brain
and face Lego with co-appeal and re-
shapeable parts." [3]

The methodology of neural modular
architecture is supported by Grand who
proposes that: "Nature, then, seems to

Baby X

The Effect of Gender Labels on Adult Responses to Infants¹

Carol A. Seavey and Phyllis A. Katz²
Graduate School and University Center,
City University of New York

Sue Rosenberg Zalk
Hunter College,
City University of New York

The present study investigated adult behavior while interacting with a three-month-old infant under conditions in which the child was introduced as a boy, as a girl, or with no gender information given. Gender labels did not elicit simple effects, but rather interacted significantly with the sex of the subject on both toy usage and physical contact measures. There was a stronger tendency for both male and female adults to utilize sex-stereotyped toys when the child was introduced as a girl. Most of the findings, however, reflected a differential response of men and women to the absence of gender information. In this condition, male subjects employed a neutral toy most frequently and handled the child least; in contrast, females used more stereotyped toys and handled the child more. All subjects attempted to guess the gender of the child (with "boy" guesses more frequent, although the child was actually female) and all justified their guess on the basis of stereotyped behavioral or physical cues like strength or softness.

A child's gender significantly affects interacting adults along many dimensions. Substantial evidence exists in both the psychological and popular literature that boys and girls are responded to in different ways, ranging from the particular types of personality traits and skills which are encouraged or discouraged, to the kinds of play and reading materials that are provided by parents and teachers (Bandura, 1960; Sears, Rau & Alpert, 1965; Mischel, 1970). The most common interpretation of such differences is that they reflect differential societal expectations as to what constitutes appropriate masculine and feminine behavior.

¹ This research was supported by Grant No. OCD-CB-151 from the Office of Child Development, Phyllis A. Katz, Principal Investigator.

² Requests for reprints should be sent to Phyllis A. Katz, Graduate Center, City University of New York, 33 West 42 Street, New York, N.Y. 10036.

<https://www.soulmachines.com/resources/research/baby-x/>

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Online



★ [This Freaky Baby Could Be the Future of AI. Watch It in Action - YouTube](https://www.youtube.com/watch?v=DycAxNQs2IU&pp=QAFIAG%3D%3D)

<https://www.youtube.com/watch?v=DycAxNQs2IU&pp=QAFIAG%3D%3D>

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Baby X is the new face of A.I.

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Baby X incorporates Face Mapping
Cognitive Neuroscience